Lost World: Looking for Anomalous Tracks in Long-term Surveillance Videos

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ABSTRACT

Video surveillance over a long span of time is no longer a luxury in this day and age. The abundance of video data captured over time presents a slew of new problems in computer vision. One potential challenge involves the task of finding anomalous tracks over a long period of time. In this work, we propose a new time-scale framework for mining anomalous track patterns in long-term surveillance videos. Track clustering is performed at two separate temporal levels to better represent the common modes of behaviour. A probabilistic anomaly prediction algorithm is also introduced to evaluate the abnormality of new tracks. In our preliminary work, experiments conducted on the LOST dataset offer insights into how track anomalies can be mined and classified. We hope this work will provide the impetus for further advancements in this direction.

Categories and Subject Descriptors

I.4 [**Image Processing and Computer Vision**]: Miscellaneous; I.2.10 [**Vision and Scene Understanding**]: Video Analysis—*video surveillance, anomaly detection*

Keywords

long-term video surveillance, track clustering, anomaly mining, video data mining

1. INTRODUCTION

As camera technology continues to advance with higher frame rates and higher resolutions, video surveillance has become increasingly pervasive and ubiquitous. Albeit the high surge of CCTV camera availability in public places, it was reported that up to 97% of surveillance videos remain unutilised. This has unseemingly created a "big data" problem [12] where far more data is collected than analysed or processed. This abundant resource can be mined to extract essential information about human activity patterns. Long term observation of scenes can potentially have far-reaching impact towards urban planning, security management and traffic monitoring.

The Long-term Observation of Scenes (with Tracks) or LOST dataset [1] is a recently established, publicly accessible dataset, comprising of 24 streaming outdoor webcams[1] captured at the same half hour each day, for over a year. It amounts to more than 6,000 hours of video, and 720 GB of compressed data. Uniquely, LOST also contains rich metadata in the form of geolocation, weather annotation, detected moving objects and object tracks or trajectories.

In long-term data, the captured tracks correspond to the change of activities across time at varying degrees; some scenes change dramatically in a short period of time (e.g. street festivals, strikes) while some changes are long-term or permanent (e.g. traffic diversions, construction). One of the immediate challenges involve the task of finding anomalous tracks over a long period of time. While abnormal event or anomaly detection is an active research area in the last decade, but almost no work has been done on surveillance videos captured on a long term basis.

In this paper, we propose a new time-scale framework for mining anomalous track patterns in long-term surveillance videos. The unlabelled track patterns are first clustered automatically through two temporal levels to find common modes of behaviour, characterised by track exemplars. A probabilistic anomaly prediction algorithm is then applied to evaluate the abnormality of a new track. Finally, we determine a track to be an anomaly if it garners sufficient confidence across all time-scales. Experiments conducted on selected cameras from the LOST dataset demonstrated the promising capability of our methods in classifying unlabelled tracks and synthetically injected anomalous tracks.

2. RELATED WORK

Many researchers have tried to find abnormal events or abnormal tracks through modeling what is typical or what is abnormal in a video scene.

In a classic paper related to object tracking and typical and abnormal event detection, Stauffer et al.[13] successfully tracked millions of objects throughout many months of video from a single location. A hierarchical classification technique was utilised, creating a codebook of motion flows to learn the motions that tend to occur together, and then unusual events are subsequently detected. However, methods that rely heavily on background modeling are known to be highly sensitive to background noise and illumination changes.

[1]The LOST dataset continues to collect to this day, but only 6 webcams are actively streaming at time of writing.

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Oh et al. [9] proposed a simple real-time framework to mine for abnormal events in indoor surveillance video streams. Instead of object tracking, they segmented raw surveillance videos into sequences of frames containing motion and calculated the total motion matrix for each segment using foreground pixel comparisons. Using multilevel hierarchical clustering and closeness-to-neighbor metric, new segments are classified into two groups: typical and abnormal, by computing their degree of abnormality. Though simplistic in idea, this work was performed on a rather low resolution video in a relatively simple indoor scene of limited variability.

Some works [10, 8] approach this problem by analysing object trajectories in video. Patterns can be extracted from trajectory data to understand the motion behaviour of objects in the video scene, and it is often necessary to perform clustering to group trajectories that are of similar nature. In a totally different work by Jiang et al. [6], an interesting approach towards anomaly detection was introduced by utilising three levels of spatiotemporal contexts to detect different forms of anomalies (point, sequential and object co-occurrence). Experiments to detect anomalies in real traffic video were easily checked against traffic rules and manual labeling of ground truth.

While most video surveillance research in anomaly detection is done on video data captured over a short span of time (from hours [9] to a few months at most [13, 16]), there are no known works that involved massively long-term video that span years. There are no other reported works in the computer vision community to date that explores this challenging new research area of long-term video surveillance, up until the initiation of the LOST dataset [1]. The originators of this dataset implemented a combination of techniques to obtain behavioural patterns for statistical and correlation analysis. Background modeling was used to detect moving objects (blobs) and formulate trajectories (tracks), while clustering is applied to find common modes of behaviour from track densities. While the paper included some general analysis of track patterns, it did not focus on detecting anomalous events from track patterns.

A recent work by Zen et al. [16] dealt with reasonably large amounts of video data, capturing the traffic flow of a whole month in New York City (though only a month but the stream ran for 24 hours each day). Typical patterns and anomalous events were discovered by an anomaly score, which defines the distance between observations taken at a specific day of the week and time of the day. The anomaly score relies heavily on accurate extraction of foreground object pixels. Also, some of the anomalous traffic flow were manually explained with hindsight knowledge of specific causal events such as Christmas Day morning, pedestrian demonstration.

Even with ground truth data of abnormal tracks, different researchers often have different definitions of what constitutes a typical or anomalous track since this largely depends on their analysis objective and track representation.

3. FRAMEWORK

The single most pressing issue with finding anomalies in long-term video surveillance lies in the lack of ground truth labels, and the sheer difficulty in establishing them correctly. In conventional anomaly detection for surveillance, ground truths are easy to establish by an expert/observer with the foreknowledge of what is expected to be labelled as an anomaly, *e.g.* illegal vehicle/traffic manoeuvres [4], a certain human behaviour or event [11], deviation from normal events in crowded scenes [7]. In the case of the LOST dataset, its structure and setup appeals to a wide range of usage such as video mining and scene analyses; more so than anomaly detection whereby ground truth labeling can be a highly arduous task considering the span of videos and the subjectivity in locating anomalies.

In this paper, we propose a novel time-scale based framework for anomaly mining[2] of track patterns in long-term surveillance video. A *time-scale* defines a temporal window that spans a specific number of days (ω) at a specific stride (ϵ) or periodicity. For instance, a weekly time-scale with a daily stride will consider all days within the last 7 days, while a monthly time-scale with a weekly stride will consider all similar days (e.g. Mondays) within the last 30 days. In our framework, a pre-defined S number of time-scales are selected to verify a potentially anomalous track. Hypothetically, the co-occurence of an anomalous track across a majority of time-scales should provide sufficient weight towards classification of tracks. The block diagram in Figure 1 shows our time-scale framework the use of several time-scales to determine which tracks are anomalous.

4. METHODS

4.1 Track Clustering

In view of the time-shifting framework for anomaly prediction, we need to increase the efficiency and effectiveness of the clustering step. Hence, we proposed an improved two-step track clustering algorithm that is able to firstly reduce the number of tracks per day when a longer period is used for learning (over-clustering step), and then optimally cluster the remaining tracks based on statistical heuristics (grouping step).

Each track T is represented as

$$t_i = \{x_i, y_i, u_i, v_i\} \quad \forall i \in 1 \ldots |T| \tag{1}$$

where x and y are the track positions while u and v are the track velocities (along the x and y directions) at frame i. Similar to the original work [1], we utilise the Chamfer distance metric to effectively capture the similarity of one track to another in an ordered, assymetrical manner. Concisely, the Chamfer distance (mean of minimum distances) D from track P to track Q is given as:

$$D(P,Q) = \frac{1}{|P|} \sum_{t_p \in P} \min_{t_q \in Q} |t_p - t_q|^2 \tag{2}$$

Assuming there are n number of tracks altogether, similarities corresponding to a $n \times n$ asymmetric pairwise distance matrix D can be represented by a Gaussian affinity matrix

$$A(P,Q) = e^{-D(P,Q)/\sigma^2} \tag{3}$$

Clustering by affinity propagation [3] is then performed on the affinity matrix A.

There are several sub-problems that need to be solved. The authors in [1] did not provide sufficient details on how the "over-clustering" step is applied to reduce the number

[2]We prefer the term *anomaly mining* instead of anomaly detection as the lack of ground truth verification constrains this to a search/mining task.

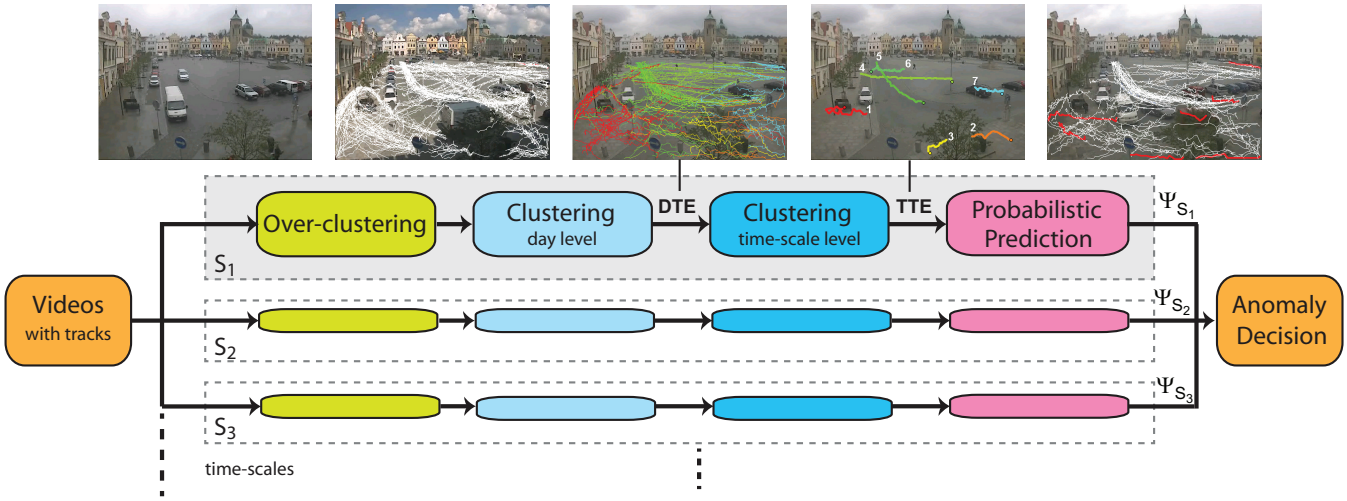

Figure 1: Proposed time-scale based framework for track anomaly mining in long-term surveillance video.

of tracks. Also, it was suggested that the preference vector for affinity propagation be assigned to the row-wise median values divided by the number of tracks represented by each over-clustered track or *children tracks*[3]. This appears to be counter-effective as it penalises tracks that are well-represented (more prominent mode of behavior) by affixing a smaller preference value. Practically speaking, this will not work well in our time-scale framework since the number of children tracks will depend on the time-scale period, *i.e.* longer periods will result in more tracks learnt, thus more children tracks per cluster.

4.1.1 Over-clustering

We take cue from an interesting variant of the k-means clustering algorithm called rek-means [2] that performs an over-clustering step to identify and remove outliers or data samples that are far from the true cluster centers. In our work, we first compute k-means on n tracks with $k = n/\rho$ number of clusters. Clusters that contained less than ρ members tend to be isolated groups that are likely to be far from a true cluster center that best describes a common mode of behavior. To balance between computational cost and information loss, we choose $\rho = 4$. At this juncture, the number of discarded tracks associated to each over-clustered track is stored.

4.1.2 Affinity propagation

Following that, we perform affinity propagation [3] on two temporal levels. Firstly, we apply it on the set of over-clustered tracks with median values as the preference vector. This results in $n_{\mathrm{DX}}(\tau)$ number of *daily track exemplars* (DTE), each representing $n_{\tau,k}$ number of original tracks in the data. Subsequently, all the DTEs within the period defined by the time-scale are combined and we perform a second round of clustering to obtain $n_{\mathrm{TX}}(s)$ number of *time-scale track exemplars* (TTE). The preference vector used this time is defined by

$$F = \tilde{A}\frac{n_{\tau,k}}{\bar{n}_{\tau,k}\alpha} \tag{4}$$

[3]By going down the cluster tree formed by hierarchical k-means, we can find all the children nodes (tracks) that are represented by a single parent node (track).

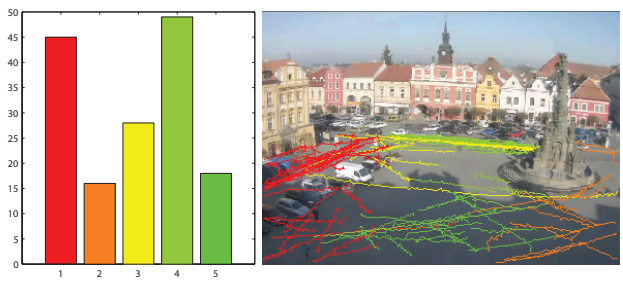

Figure 2: Histogram of clusters extracted by time-scale level clustering (left) and distribution of DTEs belonging to these clusters (right)

where the row-wise median values of the affinity matrix, $\tilde{A}$ is multiplied by the mean-normalised number of original tracks represented by their corresponding DTEs. This allows the preference values to be statistically dependent on the distribution of original tracks represented by each DTE. A tunable free parameter α further scales the preferences, enabling the generation of different number of clusters. Fig. 2 shows the histogram or densities of five clusters extracted by time-scale level clustering from the sample scene. Notice from the track lines how the daily track exemplars (DTEs) are clustered based on both position and velocity. Clusters 3 (yellow) and 4 (light green) occupy the same street but they represent opposing directions.

We devise a heuristical criterion to obtain a near-optimal value for α. Two measures are used to formulate the criterion: (i) the commonly used Tightness and Separation Criterion (TSC) [15]:

$$TSC(X,K) = \frac{\frac{1}{M}\sum_{j=1}^{K}\sum_{\ell=1}^{M} R_{\ell j}^2 |T_\ell - X_j|^2}{\min_{j_1,j_2}|X_{j1} - X_{j2}|^2} \tag{5}$$

which measures the quality of clustering (For feasibility being a non-fuzzy method, we discard the degree of membership R term.); (ii) difference between the second order derivatives of TSC and number of clusters, $\Delta_\alpha = |TSC''(\alpha) -$

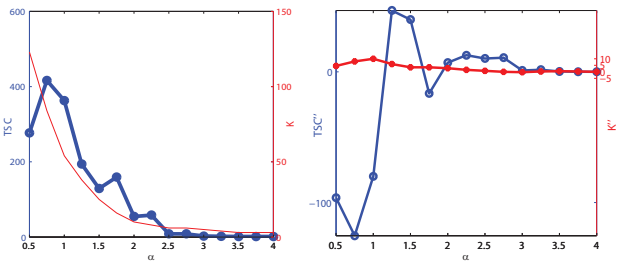

Figure 3: TSC and K plots with TSC fulfilling R1 at $\alpha = 2.5$ (left); TSC$''$ and K$''$ plots (right), with the difference value (convergence) fulfilling R2 at $\alpha = 3.0$ (right).

$K''(\alpha)|$. The criterion for selecting α is succintly governed by the following rules:

$$\mathbf{R1} : \min_{\alpha}(\text{TSC}_{\alpha} < \phi * \max(TSC)\ ) \tag{6}$$

$$\mathbf{R2} : \min_{\alpha}(\Delta_{\alpha} < \phi * \max(\Delta) \tag{7}$$

$$\alpha^* = \arg\max_{\alpha}(\mathbf{R1}, \mathbf{R2}) \tag{8}$$

where ϕ is fixed at 0.01 or 1%, a reasonable level of threshold for R1 and a good convergence point (local minimas) for R2. We evaluate the criterion in Eqs. 6-8 by clustering with a range of different α values at a fixed interval of 0.25, *i.e.* $\alpha = \{0.5, 0.75, \ldots, 4.0\}$. The plots in Fig. 3 depict how rules R1 and R2 are fulfilled, thereafter obtaining a final value of $\alpha^* = 3.0$. by Eq. 8.

4.2 Probabilistic Anomaly Prediction

Motivated by the statistical scheme presented by Hu et al. [5], we proposed a probabilistic anomaly prediction algorithm that evaluates the abnormality of a track. In our case, the exemplar tracks discovered in the clustering step characterises the important motion patterns learnt from a particular period of tracks.

By formulating the distance metric from Eq. 2 as a random variable, the likelihood probability of a new track T' given the j-th exemplar track X_j is estimated as

$$P(T'|X_j) = e^{-\eta_j D(T', X_j)} \tag{9}$$

where the parameter η_j can be estimated (based on maximal likelihood evaluation) as the reciprocal of mean distances learnt from the distribution of training track samples,

$$\eta_j = \frac{K}{\sum_{k=1}^{K} D(T_k, X_j)} \tag{10}$$

A new track T' is classified as an anomalous track if the exemplar track that possesses the maximum likelihood,

$$j^* = \arg\max_{j} P(T'|X_j) \tag{11}$$

is found to be less than a threshold γ_j^*. For ease of computation in the evaluation step, we can compute the entire set of thresholds, $\Gamma = \{\gamma_1, \gamma_2, \ldots, \gamma_C\}$ during the training step. The threshold value for the j-th exemplar track can be calculated by taking the minimum likelihood of all training samples $T_{j,\ell}$ belonging to the j-th cluster:

$$\gamma_j = \min_{\ell} P(T_{j,\ell}|X_j) \tag{12}$$

4.3 Classifying Anomalous Tracks

Owing to the lack of ground truth in long-term videos, we formulate a multiple time-scale validation scheme to further verify the anomalies found. Each track is evaluated for S number of times (corresponding to time-scales). Each anomalous track is assigned to a probability value that represents the degree of abnormality, or *anomaly confidence*.

Intuitively, the anomaly confidence score should decrease as the temporal distance (day-unit) from the present (evaluated) day increases, and vice versa. This allows an anomalous track candidate to be measured with an anomaly confidence proportional to the day of its matched exemplar track (from Eq. 11). In other words, anomalous tracks that are matched to track exemplars that are farther in the past will have less prominence. To create a non-linear "drop-off" effect, we use a sigmoid function to ensure that the anomaly confidence Ψ of candidate track T^* lies in the range of $[0.5, 1]$:

$$\Psi_{T^*} = \begin{cases} 1/(1 + exp[\beta/\delta(j^*)]) & \text{if anomalous track} \\ 0 & \text{if typical track} \end{cases} \tag{13}$$

where β is a scaling constant and $\delta(j^*)$ is the temporal distance (in unit days) of the matched exemplar track. For completeness, we assign a confidence value of 0 for tracks that are typical. This is essential as a candidate track may be anomalous on one time-scale but not on another.

Finally, a candidate track T^* is classified as an anomalous track by simply taking the mean anomaly confidence scores $\hat{\Psi}_{T^*}$ across all S time-scales, and threshold it against the lower bound of the anomaly confidence, *i.e.* 0.5. If $\hat{\Psi}_{T^*} > 0.5$, then track T^* is an anomalous track. For a more simplistic alternative, one could also employ a simple majority vote across all time-scales.

Figures 4(a)-(c) show the candidate anomalies found at different time-scales. Notice how the two tracks on the right side of the scene (behind fountain) in Fig. 4(d), are eventually classified as the anomalous tracks after taking the anomaly confidences from all time-scales into consideration. Here, our method intuitively establishes an anomaly by verifying its "confidence" of co-occurrence throughout different time-scales.

5. EXPERIMENTS

Due to the massive amount of video data available from the LOST dataset [1], we utilise only two of the webcams (that are currently still active on a daily basis) to test our proposed method—**camera '001'** (Ressel Square, Chrudim, Czech Republic) and **camera '017'** (Havlickuv Brod, Czech Republic). Camera '001' is set in an open town plaza where most moving objects are pedestrians, while camera '017' is set in a busy junction facing a carpark where most moving objects are vehicles.

In our experiments, we ad-hocly chose 3 different time-scales (each denoted by day-stride pair (ω, ϵ)): $S_1 = (28, 7)$, $S_2 = (7, 1)$, $S_3 = (30, 1)$ to learn the common mode of track behaviours. Evaluation is performed by considering each day's tracks as new "unseen" tracks whilst using past tracks from all the days defined within each time-scale to learn its respective patterns. For instance, to evaluate new tracks from day τ_i on time-scale S_1, the track patterns from the 7 previous days $\{\tau_{i-7}, \ldots, \tau_{i-1}\}$ are used for learning.

We conducted two experiments to evaluate our proposed

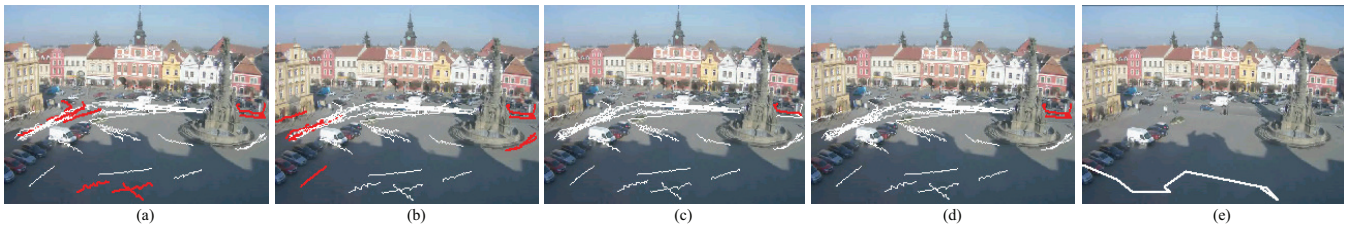

Figure 4: Sample scene from camera '001'. Candidate anomalies found at different time-scales: (a) $S_1 = (28, 7)$, (b) $S_2 = (7, 1)$, (c) $S_3 = (30, 1)$, (d) final anomalous tracks, (e) a synthetically injected anomalous track (thick white line)

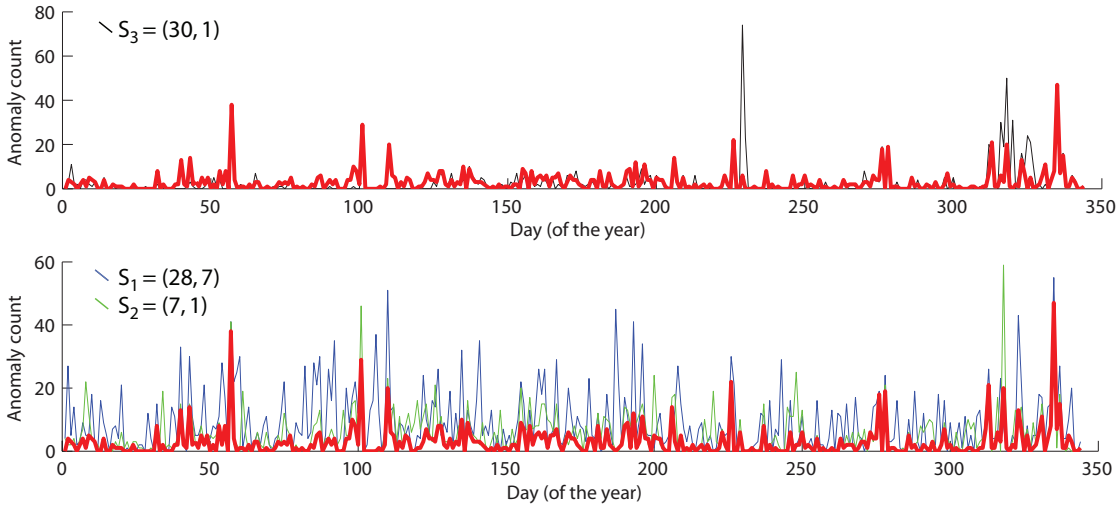

Figure 5: Breakdown of number of anomalous tracks for each day of the year (for only one selected year) for camera '001'. The red line is the final number of anomalous tracks considering all three time-scales.

technique: (i) anomaly mining on unlabelled tracks, (ii) classification of synthetically injected anomalous tracks.

5.1 Anomaly Mining

The limitations of the LOST dataset posed a challenging problem of determining which of the unlabeled tracks are anomalous. Unless a human expert is tasked to identify these anomalies (with specific description as guidance), any accuracy measures remain difficult to validate. As such, in this preliminary work, we perform anomaly mining using our proposed method. Table 1 shows that about 3.5% of all available tracks on both cameras are considered to be anomalies. In several cases, we observe that pedestrians that walk against the usual direction of flow (camera '001') and vehicles that pull over on the busy road (camera '017') are often classified as anomalous behaviours.

To examine closer the contribution or *influence* at different time-scales, we re-ran our experiment using each individual time-scale to determine the anomalies found. In Fig. 5, we observe that the S_1 and S_2 time-scales recall too many anomalies, more than what was eventually chosen (thick red line) while S_3 appears to be more stringent. This is not surprising as learning the DTEs from 30 past days is probably too inclusive that it rejects many possible anomalies. Thus the usage of all three time-scales strikes a good balance between weekly (S_1), short term (S_2) and long term (S_3) patterns. The year-end period (November to December) seemed to show a high occurence of anomalies. By manual inspection, we found that weather was unpredictable during the winter, and there were many events that took place at the plaza.

5.2 Synthetic Anomaly Injection

To deal with the lack of ground truth labels, we are inspired by a recent work by Twitter Inc.[14], on long-term anomaly detection given the somewhat similar nature of their data. In our case, a single anomalous track is synthetically generated and injected into the set of test tracks for each evaluated day τ_i. The injected anomalies are created by randomly assigning a starting point (at an unused position), followed by a path (of average length) with a velocity fluctuation of around 2.0σ, 2.5σ and 3.0σ (standard deviation) of a Gaussian distribution. The feature vectors (Eq. 1) of the typical tracks were used to fit within 1.0σ of the distribution. Fig. 4(e) shows a sample synthetically injected anomalous track that was generated for classification.

Table 1 shows the classification results on both cameras, with different values of σ. Overall, we are able to reach a top classification rate of 93.0% when anomalies are injected at 3.0σ of the track velocity distribution.

6. CONCLUSIONS

In this paper, we have proposed a new time-scale framework for finding anomalous tracks in long-term surveillance

Table 1: Results of anomaly mining and classification of synthetic anomalies

Camera	Mined anomalies (%)	Classification rate (%)		
		$\sigma = 2.0$	$\sigma = 2.5$	$\sigma = 3.0$
001	3.56 (1012/28407)	67.2	82.6	93.0
017	3.54 (1243/35100)	22.9	58.3	76.7

video—an upcoming area in computer vision with much potential. Various sub-problems within the framework were addressed, notably the handling a large number of tracks by clustering at two temporal levels, and a probabilistic anomaly prediction algorithm for evaluating the abnormality of a new track. Some preliminary results on two sets of experiments were discussed and analysed.

Future Directions. There remains numerous open problems in this new research. Most essentially, the lack of anomaly ground truth hampers the feasibility of performing detection. Even if an enormous effort is put into labelling, how accurate can an expert pinpoint an anomalous behaviour through a long timeframe? Is there an effective mechanism can enables the verification of these mined anomalies? We intend to run further experiments on more cameras, and to investigate how trend line patterns [16] can be used to authenticate a predicted anomalous track.

7. ACKNOWLEDGMENTS

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